Beyond Answer Attribution: Recovering Supporting Evidence in RAG

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Abstract

In retrieval-augmented generation (RAG), identifying which source passages contribute to a generated response is important for evidence attribution and citation. Existing approaches rely on textual similarity, attention weights, counterfactual inference, or direct logit attribution. However, methods that focus on direct contributions to the response may overlook supporting evidence, particularly in multi-hop question answering. We investigate how the choice of attribution target affects evidence ranking, distinguishing multi-hop supporting-evidence recovery from answer localization in more extractive question-answering tasks. In particular, we examine whether attribution to question tokens can complement response attribution in recovering supporting evidence. To evaluate this, we build on the Logit attribution method of Ferrando et al. (2023), which decomposes attention outputs into source-specific writes. We introduce GROUNDHOG (Head Output Grounding), a training-free method that combines normalized logit attributions to question and response tokens. Across multiple language models, we find that combining both targets improves recovery of human-annotated supporting evidence in multi-hop question answering, while response-only attribution remains stronger for simple answer localization.

1 Introduction

Retrieval-augmented generation (RAG) conditions language model responses on external sources, making it important to identify which passages provide evidence for a generated answer (Lewis et al., 2020). This is particularly challenging in multi-hop question answering, where supporting evidence may be distributed across several passages. While some passages contain information directly associated with the answer, others establish intermediate relationships needed to support it. Consequently, identifying the source most closely associated with an answer is not necessarily sufficient to recover the broader set of supporting evidence.

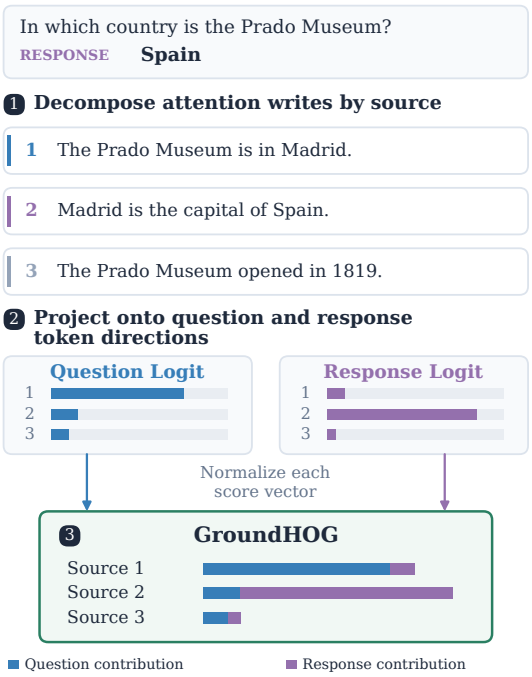


Figure 1: **GROUNDHOG** enables evidence localization by projecting source-specific attention writes onto the output-token directions of the question and response, then aggregating the resulting source scores.

Existing source-attribution approaches estimate passage importance through textual similarity, entailment, context perturbations, or signals extracted from the generating model (Gao et al., 2023; Cohen-Wang et al., 2024; Qi et al., 2024). Model-internal approaches offer a way to examine how source information contributes to predictions. In particular, attention-based methods estimate source influence from the model's attention patterns, while Logit attribution (Ferrando et al., 2023) decomposes attention outputs into source-specific contributions to predicted token logits. However, when attribution is evaluated only with respect to response tokens, it primarily measures direct con-

tributions to the answer prediction. Passages establishing intermediate relationships may receive relatively little attribution even when they are annotated as necessary supporting evidence.

We investigate whether attribution to question tokens provides a complementary signal for identifying supporting passages. Question-side attribution measures how source-specific attention writes contribute to predictions of the question tokens, potentially emphasizing information relevant to relationships expressed in the query. Because it is computed before the response, this signal reflects question relevance independent of the answer.

Building on the Logit attribution formulation of Ferrando et al. (2023), we introduce GROUNDHOG (Head Output Grounding), a lightweight training-free method that combines question- and response-targeted attribution for source ranking. GROUNDHOG aggregates signed attention-write contributions over source spans, separately normalizes the resulting question and response attribution vectors, and combines them with equal weight. This balances two signals that may emphasize different aspects of source relevance without requiring additional training or repeated context perturbations.

We evaluate GROUNDHOG on four question-answering benchmarks, distinguishing multi-hop supporting-evidence recovery from answer localization. Across multiple language models, combining separately normalized question and response attribution consistently improves average precision over response-only Logit attribution on the multi-hop benchmarks. In contrast, response-only attribution remains stronger for answer localization. Ablations show that both normalization and attribution target selection contribute to the observed improvements. Additional comparisons examine whether these effects extend to attribution methods that propagate contributions to input tokens. We also report exploratory hallucination-detection experiments with HOGWASH in Appendix A.

Together, these findings show that attribution to answer predictions can provide a strong signal for locating answer-associated passages, but may not recover all annotated supporting evidence. Incorporating question-targeted attribution offers a simple way to improve multi-hop evidence recovery.

2 Background and Related Work

Source attribution in RAG aims to identify which parts of the retrieved context are relevant to a generated response. Existing approaches differ in whether they assess the relationship between source and response texts or examine the sources' influence on the generating model. This distinction is sometimes framed as *corroborative* versus *contributive* attribution (Worledge et al., 2024). While corroborative attribution identifies evidence supporting a generated statement, contributive attribution seeks to explain which inputs influenced the model's prediction. Wallat et al. (2025) further distinguish citation correctness from citation faithfulness, showing that a citation can support an answer without reflecting the model's actual reliance on the cited document. We review methods from both perspectives, focusing on their application to source attribution in question answering.

2.1 Textual Similarity Methods

Text-based approaches identify relevant source passages by comparing them with the question or generated response. Lexical methods such as BM25 rank passages through term matching, while embedding-based methods compare learned semantic representations (Robertson and Zaragoza, 2009; Reimers and Gurevych, 2019). These approaches can identify passages related to the answer, although similarity alone does not establish whether a passage supports a particular claim. Natural language inference (NLI) methods address this distinction by estimating whether the information in a passage entails the generated statement, as in the citation evaluation framework of Gao et al. (2023). LLM-based judges can similarly assess whether a candidate passage provides sufficient support for an answer. Saxena et al. (2025) compare generation-time and post-hoc citation methods, finding trade-offs between citation coverage and correctness.

These methods operate only on model input and output without requiring access to the generating model. They are therefore useful for identifying supporting evidence, but do not directly establish whether the model relied on that evidence when generating its response.

2.2 Context-Perturbation Attribution

Context-perturbation methods instead estimate the influence of a source by modifying the model's input and observing changes in its predictions. Leave-

one-out attribution removes individual passages and measures the resulting change in the likelihood of the response. ContextCite (Cohen-Wang et al., 2024) extends this idea by fitting a sparse linear model to the effects of randomized context ablations, allowing source contributions to be estimated from a set of perturbation experiments. AttriBoT (Liu et al., 2025) reduces the computational cost of approximating leave-one-out attribution through cached activations, hierarchical attribution, and smaller proxy models.

By measuring changes in model behavior, these methods can capture effects that are not apparent from textual similarity or individual model components. However, they generally require repeated evaluations under modified contexts, which can be computationally expensive. Removing passages also changes the input distribution and may affect predictions through changes in context length, token positions, or interactions among the remaining sources.

2.3 Model-Internal Attribution

Model-internal approaches estimate source contributions using the representations and computations of the generating model. Gradient-based attribution methods, such as Integrated Gradients (Sundararajan et al., 2017), estimate input importance from the sensitivity of model predictions to input representations. In the RAG setting, MIRAGE (Qi et al., 2024) uses model-internal saliency to identify context-sensitive answer tokens and associate them with influential source tokens. Other approaches use hidden-state representations or similarities to identify relevant evidence, avoiding repeated context-removal experiments. Sun et al. (2026) introduce a controlled evaluation framework for context-utilization explanations, assessing whether token-level highlights identify context information actually used by a model.

In-context re-ranking (ICR) uses attention from query tokens to preceding documents to estimate document relevance without generating a response (Chen et al., 2025). It calibrates these scores using a content-free query and evaluates retrieval on both single-hop and multi-hop benchmarks. Like ICR, our question-side attribution extracts a relevance signal while processing the query. However, GroundHOG measures signed attention-write contributions to token logits rather than attention weights, and combines question-side scores with response attribution to recover supporting evidence.

Attention-based attribution. Attention provides a more direct connection between generated tokens and preceding context positions. Several approaches aggregate attention weights to estimate the relevance of source passages, sometimes incorporating additional structural information, such as dependency relationships. However, attention weights describe how strongly a position attends to another without accounting for the content of the value vectors or their transformation through the output projection (Jain and Wallace, 2019; Kobayashi et al., 2020).

Methods based on attention contributions address this limitation. Ferrando et al. (2023) introduce Logit attribution, which decomposes attention outputs into position-specific contributions and projects them onto the predicted token's unembedding direction. Their ALTI-Logit extension additionally accounts for contextual mixing across layers. Value Zeroing (Mohebbi et al., 2023) takes a different approach, measuring the effect of suppressing individual value vectors on intermediate representations. More recently, Gema et al. (2026) use output-value circuit contributions to identify attention heads involved in non-literal retrieval, contrasting contributions from relevant and irrelevant context positions and validating the selected heads through ablation.

Our work builds on the Logit formulation of Ferrando et al. (2023). Rather than introducing a new decomposition of transformer computations or identifying retrieval heads, we investigate how the choice of attribution target affects supporting-evidence recovery in RAG. We evaluate attribution to question and response predictions separately and in combination, using GROUNDHOG to balance their contributions to source evidence scores.

3 Attention Attribution for Supporting-Evidence Identification

Consider a language model answering a query using retrieved sources. To identify which sources contribute to its predictions, we use the Logit attribution method of Ferrando et al. (2023), which decomposes attention outputs into source-specific writes and measures their contributions to predicted token logits. We investigate how these contributions can be aggregated over different prediction targets to identify supporting evidence in RAG.

Source-specific write attribution. Consider a causal language model predicting token y_{t+1} from

the residual state at position t. At layer ℓ and head h, let $\alpha_{t,j}^{\ell,h}$ denote the post-softmax attention weight from position t to position $j \le t$, and let $v_j^{\ell,h}$ be its corresponding value vector. The head's contribution to the residual stream at position t is

$$w_t^{\ell,h} = W_O^{\ell,h} \sum_{j=0}^{t} \alpha_{t,j}^{\ell,h} v_j^{\ell,h}, \tag{1}$$

where $W_O^{\ell,h}$ is the head's slice of the output projection. The contribution from position j to this attention write is therefore

$$w_{t,j}^{\ell,h} = W_O^{\ell,h} \alpha_{t,j}^{\ell,h} v_j^{\ell,h}. \tag{2}$$

We measure the contribution of each source-specific write to the logit of the next token y_{t+1} by projecting it onto the corresponding unembedding vector $U_{y_{t+1}}$, adjusted by the model's final RMSNorm:

$$a_{t,j}^{\ell,h} = \left(\frac{\gamma \odot U_{y_{t+1}}}{\rho_t} \right)^{\top} w_{t,j}^{\ell,h}, \qquad \rho_t = \sqrt{\frac{1}{d}\|x_t\|_2^2 + \varepsilon}. \tag{3}$$

Here, γ is the learned RMSNorm scale, x_t is the final residual state at position t before normalization, d is its dimensionality, and ε is the normalization constant.

Let S denote the set of retrieved source token positions, and R the set of positions t whose next-token targets y_{t+1} belong to the response. For each source token $j \in S$, we sum its attribution over layers, attention heads, and response prediction positions:

$$r_j = \sum_{\substack{t \in R \\ t \ge j}} \sum_{\ell=1}^{L} \sum_{h=1}^{H} a_{t,j}^{\ell,h}. \tag{4}$$

This measures the summed direct logit attribution of attention writes from the contextualized value state at source position j to response-token predictions. We refer to this method as Summed Logit.

3.1 GroundHOG: Question–Response Evidence Ranking

Applying Summed Logit attribution over the response positions measures the direct contributions of source tokens to response logits. However, it may overlook annotated evidence not directly reflected in the response. We therefore also consider attribution to question-token predictions, which may provide a complementary signal for identifying supporting evidence.

Let Q denote the set of positions t whose next-token targets y_{t+1} belong to the question. Analogously to the response-side attribution r_j, we compute the question-side attribution for each source token $j \in S$ as

$$q_j = \sum_{\substack{t \in Q \\ t \ge j}} \sum_{\ell=1}^{L} \sum_{h=1}^{H} a_{t,j}^{\ell,h}. \tag{5}$$

Since the retrieved sources precede the question, question-side attribution is computed before the response and is independent of the generated answer. It therefore provides a model-internal signal of source relevance to the question rather than a measure of source influence on the response.

To obtain source-level scores, we aggregate the signed token attributions within each candidate source span. Let $S_1, \ldots, S_K$ denote the sets of token positions belonging to the retrieved sources. The question-side and response-side scores for source S_k are

$$q_k = \sum_{j \in S_k} q_j, \qquad r_k = \sum_{j \in S_k} r_j. \tag{6}$$

We then normalize the source-level attribution vectors separately by their sums of absolute values and combine them with equal weight:

$$\mathrm{GroundHOG}(S_k) = \frac{q_k}{\sum_{i=1}^{K} |q_i|} + \frac{r_k}{\sum_{i=1}^{K} |r_i|}. \tag{7}$$

4 Experimental Setup

Datasets. We evaluate source attribution on four question-answering benchmarks, distinguishing multi-hop supporting-evidence recovery from answer localization. HotpotQA (Yang et al., 2018) and MuSiQue (Trivedi et al., 2022) evaluate multi-hop supporting-evidence recovery, where multiple passages may be necessary to answer a question. We use the supporting-fact annotations from HotpotQA and supporting-paragraph annotations from MuSiQue to identify relevant sources. For answer localization, we use MRQA (Fisch et al., 2019) and WikiQA (Yang et al., 2015). MRQA sources are segmented into sentences and labeled according to overlap with an annotated answer span, while WikiQA uses its human-annotated answer-sentence relevance labels. The selection procedures and statistics are detailed in Appendix B.

Models. We evaluate Llama-2-7B-Chat (Touvron et al., 2023), Llama-3.1-8B-Instruct (Grattafiori et al., 2024), Qwen3-4B-Instruct-2507 (Yang et al., 2025), and Mistral-7B-Instruct-v0.3 (Jiang et al., 2023). All source-attribution methods are evaluated using the supplied response under teacher forcing. Our prompt, shown in Appendix C, presents the retrieved sources before the question, allowing question positions to attend to source tokens.

Baselines. We compare GroundHOG against textual retrieval, model-internal attribution, and context-perturbation methods. Textual baselines include BM25 (Robertson and Zaragoza, 2009), BGE embedding similarity (Xiao et al., 2024), and DeBERTa-based natural language inference (Sentence Transformers, n.d.). These approaches rank candidate sources using the supplied question and reference response without accessing the generating model's internal representations.

Model-internal baselines include aggregated attention weights, input-gradient saliency, MIRAGE (Qi et al., 2024), and attention-based attribution methods including AttnTrace (Wang et al., 2026) and AttnLRP (Achtibat et al., 2024). ALTI-Logit (Ferrando et al., 2023) propagates contributions through earlier context mixing to the input embeddings. We additionally compare against ARC-JSD (Li et al., 2026), exact leave-one-out attribution, and ContextCite (Cohen-Wang et al., 2024), which estimate source importance from changes in model predictions under context perturbations.

To isolate the contribution of GroundHOG's ranking procedure, we evaluate Summed Logit attributions over the question only and the question and response together, which combines both attribution signals without separate normalization.

Evaluation metrics. We report precision at one (P@1) and average precision (AP), averaged over evaluation questions. P@1 measures whether the highest-ranked candidate is annotated as supporting evidence, while AP evaluates the ranking of all annotated evidence. We treat AP as the primary metric for multi-hop recovery, where identifying the broader set of supporting passages is particularly important. For MRQA, we compute each metric separately for its constituent datasets and report their macro average.

5 Results

5.1 Attribution Targets and Evidence Recovery

Question and response attribution. Table 1 demonstrates the contributions of question-side and response-side attribution across a broad set of language models. On the multi-hop benchmarks, neither signal alone matches their combination. In particular, question-only attribution performs worse in P@1 than response-only attribution, but better in HotpotQA AP and worse in MuSiQue AP, suggesting that question relevance alone is insufficient to identify the complete supporting evidence. Nevertheless, incorporating this signal substantially improves average precision when combined with response attribution.

The comparison between Summed Logit (question + response) and GroundHOG further demonstrates the importance of balancing these contributions. Summing question and response attribution without separate normalization already improves evidence recovery, but GroundHOG achieves higher mean AP. This suggests that the relative scale of the two signals is important, and that normalizing them independently allows each to contribute useful information without dominating the other.

Multi-hop supporting-evidence recovery. Table 2 compares GroundHOG with existing source-attribution methods on HotpotQA and MuSiQue. GroundHOG improves average precision over response-only Summed Logit in all six displayed dataset/model settings. It also achieves higher mean AP than the lexical, embedding, and context-perturbation baselines. However, AttnLRP with normalized question and response attribution achieves higher AP on HotpotQA across all three models. This suggests that the benefit of combining attribution targets is not specific to direct Logit attribution.

Answer localization. Table 3 shows results for MRQA and WikiQA. In contrast to multi-hop supporting-evidence recovery, response-only attribution generally provides a stronger signal when evidence is closely associated with the answer. When evidence is distributed across multiple supporting passages, incorporating question-side attribution becomes more useful.

Target / aggregation	Llama-2	Llama-3.1	Qwen3	Mistral
HotpotQA				
Question only	.634/.706	.692/.740	.679/.716	.681/.734
Response only	.619/.800	.661/.852	.621/.830	.655/.842
Question + response, unnormalized sum	.767/.814	.800/.830	.798/.826	.790/.828
GroundHOG, normalized sum	**.779/.842**	**.825/.884**	**.808/.874**	**.811/.876**
MuSiQue				
Question only	.570/.674	.626/.758	.641/.744	.625/.732
Response only	.635/**.910**	.671/.954	.655/.944	.663/.944
Question + response, unnormalized sum	.751/.870	.799/.916	.816/.924	.792/.902
GroundHOG, normalized sum	**.772**/.908	**.821/.962**	**.824/.950**	**.808/.946**

Table 1: Question and response attribution ablations, AP/P@1. Each model–dataset column uses identical examples across the four variants. Bold and underlining mark the highest and second-highest values, including ties.

Method	HotpotQA			MuSiQue		
	Llama-2	Llama-3.1	Qwen3	Llama-2	Llama-3.1	Qwen3
Text retrieval						
BM25	.698/.818	.698/.818	.698/.818	.676/.801	.676/.801	.676/.801
BGE (q+a)	.748/.829	.748/.829	.748/.829	.707/.816	.707/.816	.707/.816
Model-internal baselines						
Attention (q+a)	.469/.437	.722/.746	.530/.461	.397/.277	.628/.668	.497/.436
MIRAGE	.256/.308	.310/.436	.334/.480	.388/.467	.410/.538	.433/.575
AttnTrace-1	.497/.589	.740/.860	.436/.337	.529/.680	.751/.939	.458/.291
ALTI-Logit (response)	.604/.712	.593/.678	.575/.672	.595/.822	.621/.862	.604/.856
ALTI-Logit (normalized q+r)	.618/.626	.585/.546	.563/.554	.606/.742	.616/.752	.600/.740
AttnLRP (response)	.655/.824	.699/.868	.679/.866	.644/**.932**	.656/.958	.627/.958
AttnLRP (normalized q+r)	**.793/.858**	**.839/.900**	**.820/.896**	.736/.918	.818/.960	.779/.956
Context perturbation						
ARC-JSD	.648/.787	.738/.866	.709/.858	.647/.793	.762/.914	.750/.904
Leave-one-out	.527/.772	.454/.673	.521/.730	.576/.857	.603/.918	.665/.941
ContextCite	.475/.720	.483/.717	.541/.756	.595/.929	.619/.949	.668/**.960**
Direct Logit attribution						
Summed Logit (response)	.619/.800	.661/.852	.621/.830	.635/.910	.671/.954	.655/.944
Summed Logit (q+r)	.767/.814	.800/.830	.798/.826	.751/.870	.799/.916	.816/.924
GROUNDHOG	.779/.842	.825/.884	.808/.874	**.772**/.908	**.821/.962**	**.824**/.950

Table 2: **Multi-hop supporting-evidence recovery.** Entries report AP/P@1 (higher is better). Bold and underlining mark the highest and second-highest displayed values for each metric and model, including ties. ALTI-Logit and AttnLRP propagate attribution to input tokens. Normalized q+r separately normalizes question and response source scores before addition. MRQA and WikiQA comparisons are in Table 3; four-model direct Logit ablations are in Table 1.

Attribution targets and propagation. Applying the same separately normalized question–response combination to full-input AttnLRP increases AP on the multi-hop benchmarks and achieves the highest HotpotQA AP among the compared methods. However, this approach does not consistently improve ALTI-Logit over response-only attribution, indicating that the benefit depends on the underlying attribution method.

Qualitative evidence attribution. Figure 2 illustrates how attention, response-only Summed Logit, and GroundHOG distribute attribution over a multi-hop question involving Keith Nichol, the Oklahoma Sooners, and Norman. The methods emphasize different parts of the context: Summed Logit assigns prominent attribution to several question-related entities, while GroundHOG concentrates more strongly on the answer location. Although combining question and response attribution improves source-level evidence recovery on average, this token-level example does not by itself establish greater attribution to every intermediate supporting relationship.

Method	MRQA			WikiQA		
	Llama-2	Llama-3.1	Qwen3	Llama-2	Llama-3.1	Qwen3
Text retrieval						
BM25	.702/.577	.702/.577	.702/.577	.930/.908	.930/.908	.930/.908
BGE (q+a)	.727/.599	.727/.599	.727/.599	.951/.936	.951/.936	.951/.936
Model-internal baselines						
Attention (q+a)	.549/.382	.653/.523	.595/.445	.759/.628	.903/.849	.870/.794
MIRAGE	.528/.400	.551/.441	.603/.514	.731/.663	.785/.743	.807/.779
AttnTrace-1	.635/.461	.781/.690	.693/.572	.786/.610	.970/.972	.908/.858
ALTI-Logit (response)	.691/.558	.638/.481	.669/.534	.924/.896	.904/.846	.914/.873
ALTI-Logit (normalized q+r)	.595/.401	.556/.352	.572/.373	.840/.729	.822/.688	.824/.692
AttnLRP (response)	**.805/.715**	.803/**.714**	.795/.711	.965/.964	**.972/.986**	.966/**.977**
AttnLRP (normalized q+r)	.778/.689	.780/.696	.765/.672	.964/.964	.966/.977	.964/.968
Context perturbation						
ARC-JSD	.690/.599	.715/.635	.697/.618	.928/.922	.964/.963	.946/.940
Leave-one-out	.683/.612	.625/.553	.642/.577	.917/.936	.882/.899	.885/.899
ContextCite	.701/.623	.657/.573	.683/.602	.921/.932	.882/.893	.891/.906
Direct Logit attribution						
Summed Logit (response)	**.805**/.710	**.806**/.713	**.807/.717**	**.969/.968**	.969/.977	**.970**/.973
Summed Logit (q+r)	.710/.582	.709/.579	.728/.611	.945/.932	.958/.955	.964/.964
GROUNDHOG	.776/.681	.781/.689	.785/.698	.956/.959	.964/.977	.962/.964

Table 3: **MRQA and WikiQA answer-localization results.** Entries report AP/P@1 (higher is better). Bold and underlining mark the highest and second-highest displayed values for each metric and model, including ties. MRQA is macro-averaged across its 12 subsets. Normalized q+r separately normalizes question and response source scores before addition.

Attribution	Question	Response	Unnormalized q+r	Normalized q+r
Direct Logit	.692	.661	.800	**.825**
ALTI-Logit (to input tokens)	.470	**.593**	.530	.585
AttnLRP (to input tokens)	.738	.699	.832	**.839**

Table 4: **Target combination across attribution methods.** AP on Llama-3.1/HotpotQA. Normalized direct Logit is GroundHOG. Bold and underlining mark the highest and second-highest AP within each method. Full-input AttnLRP improves over response-only attribution; full-input ALTI does not. Further comparisons are in the appendix.

6 Discussion

Direct attribution and supporting evidence. Our source-attribution results highlight a distinction between identifying the context most directly associated with an answer and recovering the evidence needed to support it. Response-only Summed Logit attribution performs particularly well on answer localization, where the relevant evidence is often closely associated with the predicted answer. In multi-hop question answering, however, supporting evidence may establish intermediate relationships that are not directly reflected in the answer tokens. This may help explain why response-only attribution can identify a highly relevant passage while assigning relatively little importance to other necessary evidence. Incorporating question-side attribution provides an additional signal for identifying sources relevant to the broader information need. This question-side signal is independent of the response and does not establish whether the model relied on a passage when generating its answer. GroundHOG combines this relevance signal with response attribution to improve evidence ranking, rather than providing a causal explanation of the generated response. This extends the use of logit contributions beyond response-token attribution (Ferrando et al., 2023) and complements work using related contributions to identify retrieval heads (Gema et al., 2026). The stronger HotpotQA results of AttnLRP with normalized question and response attribution further indicate that attribution-target selection, rather than the direct Logit scoring formulation alone, is central to the observed evidence-recovery improve-

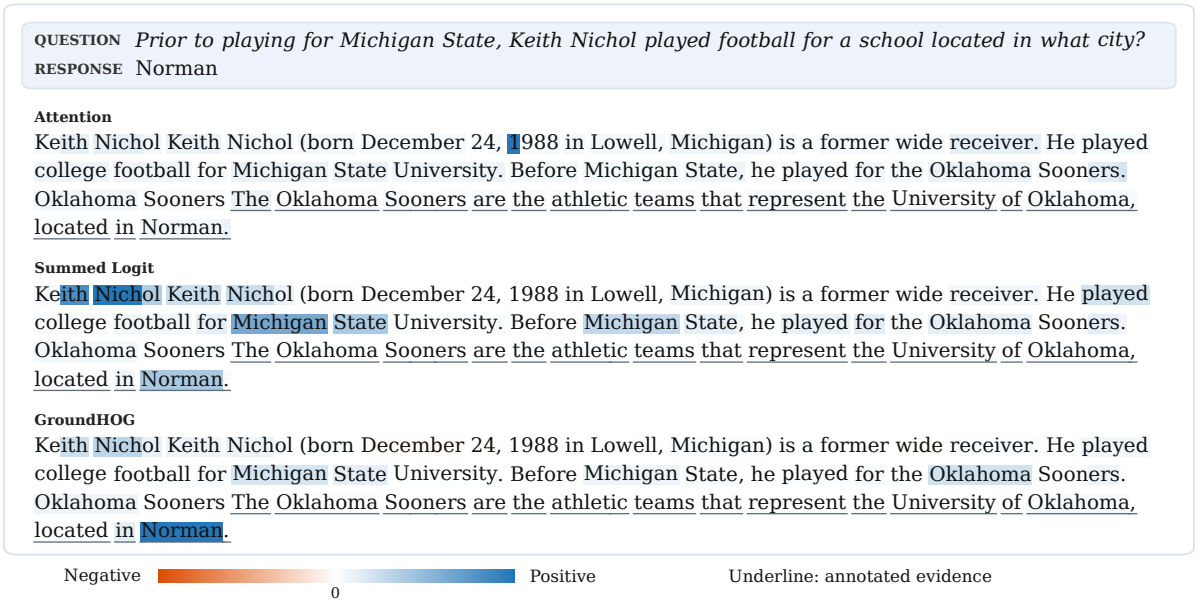


Figure 2: **Qualitative example of evidence attribution in multi-hop question answering.** Comparison of attention, Summed Logit, and GroundHOG attributions for a HotpotQA example using Llama-2. Blue indicates positive attribution and orange indicates negative attribution, with color intensity reflecting attribution magnitude. Underlined text marks human-annotated supporting evidence.

ments.

The benefit of combining question and response attribution (Table 1) is particularly interesting given that question-only attribution has lower P@1 but higher HotpotQA AP than response-only attribution. This suggests that the two signals capture different aspects of source relevance rather than providing interchangeable estimates of evidence importance. Response attribution tends to emphasize information associated with the predicted answer, while question attribution may highlight passages relevant to resolving relationships expressed in the query. The higher mean AP of separately normalized attribution over the unnormalized combination further suggests that balancing the two signals is important. Nevertheless, these results do not establish that the method explicitly reconstructs the model's reasoning process. This distinction is also relevant to evaluations of whether attribution highlights reflect actual context utilization (Sun et al., 2026).

Attribution and factual support. Computational attribution and factual grounding are related but distinct concepts. A source can strongly influence a prediction without supporting it, and a response may be factually correct even when its direct prompt-originating attribution is weak. This distinction is consistent with work separating citation correctness from faithfulness (Wallat et al., 2025) and evaluating generation-time and post-hoc citation methods (Saxena et al., 2025). Similarly, low attribution to a supporting passage does not necessarily imply that the passage was irrelevant to the model's computation, since its information may have been incorporated into other contextualized representations. GroundHOG therefore does not provide a direct measure of factual entailment. Its scores should instead be understood as model-internal signals that can assist evidence ranking.

7 Conclusion

We investigated attribution-target selection for evidence recovery in retrieval-augmented generation. Building on the Logit attribution formulation, we introduced GroundHOG, which combines question and response attribution to rank supporting passages. Across multi-hop question-answering benchmarks, GroundHOG consistently improves supporting-evidence recovery over response-only attribution, while the latter remains particularly effective for answer localization.

Limitations

Our approach relies on direct attention-write attribution rather than a complete decomposition of information flow through the model. The value

vectors associated with source positions are contextualized by earlier transformer layers and may contain information originating from other positions. Consequently, the attribution assigned to a source position does not necessarily represent the contribution of the original input token alone. GroundHOG does not explicitly trace these contributions through earlier layers, and its scores should not be interpreted as causal estimates of source influence.

The effectiveness of question–response attribution also depends on the annotation scheme and structure of the task. While GroundHOG consistently improves multi-hop supporting-evidence recovery in our experiments, response-only Summed Logit attribution remains stronger on answer localization. Combining question and response attribution is therefore not uniformly beneficial. Moreover, our multi-hop evaluation measures agreement with annotated supporting passages rather than whether the model actually followed the reasoning steps implied by those annotations.

The source-attribution experiments evaluate supplied reference responses under teacher forcing. This allows comparisons across scoring models using the same answer, but does not establish that those models would generate the corresponding responses independently. The evaluation also depends on the quality and completeness of supporting-evidence annotations. In particular, answer-overlap labels in MRQA identify answer locations rather than all information that may be useful for answering a question. Our conclusions about evidence recovery are therefore specific to the respective annotation schemes.

Finally, GroundHOG requires white-box access to the model's attention computations and value projections. Although it avoids additional training and repeated context perturbations, extracting head-level contributions can introduce substantial memory and computational overhead. Further evaluation across architectures, domains, and naturally generated responses is needed to establish the generality of these findings.

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A Hallucination Detection

We further investigate whether direct attention-write attribution can provide a useful signal for contextual hallucination detection. Whereas GroundHOG uses source-specific contributions to rank supporting evidence, hallucination detection concerns the relationship between the supplied context and the generated response as a whole. We introduce HOGWASH, a training-free detector that measures the aggregate contribution of prompt-originating attention writes to the response logits.

A.1 HOGwash: Detecting Ungrounded Responses

A response grounded in its input context may be expected to receive positive contributions from the contextual information available to the model. Conversely, a response that relies less on the supplied input may receive weaker positive contributions from prompt-originating attention writes. Motivated by this distinction, we use the direct Logit attribution of the complete prompt as a signal for contextual hallucination detection.

Let P denote the positions belonging to the input prompt, including instructions, retrieved documents, and the question, and let R denote the prediction positions corresponding to the generated response y. Using the source-specific attribution $a^{\ell,h}_{t,j}$ defined in Equation 3, we compute

$$\mathrm{HOGwash}(y) = -\frac{1}{|R|}\sum_{t\in R}\sum_{\ell=1}^{L}\sum_{h=1}^{H}\sum_{\substack{j\in P\\ j\le t}} a^{\ell,h}_{t,j}. \qquad (8)$$

The score measures the negative mean contribution of prompt-originating attention writes to the generated response logits. Higher scores indicate greater hallucination risk. Contributions remain signed, allowing attention writes that increase and decrease the target logits to offset one another. Averaging over response positions also reduces the direct dependence of the score on response length.

Unlike GroundHOG, HOGWASH does not rank individual source passages or separately normalize question and response attribution. Instead, it aggregates contributions from the entire input prompt, including information outside the retrieved documents.

The detector requires a single forward evaluation without additional training or context perturbations.

A.2 Contextual Hallucination Detection

Datasets. We evaluate HOGwash on RAGTruth (Niu et al., 2024) and HDM-Bench. RAGTruth provides annotated responses generated in retrieval-augmented settings, including question answering, summarization, and data-to-text generation. We formulate detection as a response-level binary classification problem, labeling a response as hallucinated when its annotations indicate unsupported generated content.

Unlike the source-attribution evaluation, which uses reference answers, hallucination detection operates on complete generated responses. The scoring model is not necessarily the model that originally generated the response. We preserve the

supplied context and complete response when comparing methods, ensuring that detector scores are evaluated against identical targets.

Baselines. We compare HOGwash against token entropy, negative mean log-probability, context-removal gain, RAUQ, and LLM-Check (attention). Our principal comparison is LUMINA (Yeh et al., 2026), a training-free detector that estimates hallucination risk using contextual and internal-knowledge signals. Both datasets are evaluated using Llama-2-7B-Chat as the scoring model.

Evaluation metrics. We report area under the receiver operating characteristic curve (AUROC) as the primary detection metric and area under the precision–recall curve (AUPRC) as a complementary measure. Both metrics are computed at the response level, with hallucinated responses treated as the positive class.

A.3 Results

Hallucination detection results. Table 5 compares HOGwash with uncertainty-based and attribution-based baselines on RAGTruth and HDM-Bench. On RAGTruth, HOGwash achieves .759 AUROC and .740 AUPRC, compared with .744 and .731 for LUMINA. On HDM-Bench, HOGwash achieves .773 AUROC and .885 AUPRC, compared with .734 and .847 for LUMINA. HOGwash obtains the highest AUROC and AUPRC among the displayed methods on both datasets.

Limitations. HOGwash measures contributions from the complete input prompt, including instructions and questions as well as retrieved documents. Its score does not isolate the contribution of retrieved evidence alone or identify which individual claims in a response are hallucinated. Moreover, prompt dependence does not necessarily imply factual support. We do not report statistical significance tests for the updated comparisons.

B Datasets and Evaluation Protocols

B.1 Source Attribution Datasets

We use HotpotQA Distractor development data, MuSiQue-Answerable development data, MRQA validation and test data, and the official WikiQA test split. HotpotQA candidates are sentences with supporting-fact labels, while MuSiQue candidates are paragraphs with supporting-paragraph labels. MRQA candidates are sentences labeled by overlap with an annotated answer span, and WikiQA uses human-annotated answer-sentence relevance labels. The evaluation includes 500 HotpotQA questions, 500 MuSiQue questions, 1,148 MRQA questions, and 221 WikiQA questions.

B.2 Hallucination Detection Datasets

We evaluate hallucination detection on RAGTruth and HDM-Bench using Llama-2-7B-Chat as the scoring model. The RAGTruth evaluation includes 450 responses, and the HDM-Bench evaluation includes 190 responses.

B.3 Evaluation Metrics and Statistical Testing

For source attribution, we report precision at one (P@1) and average precision (AP), using expected scores under uniform tie-breaking. HotpotQA, MuSiQue, and WikiQA average metrics across questions, while MRQA uses macro-averages across its constituent datasets.

For hallucination detection, we report AUROC and AUPRC, treating hallucinated responses as the positive class. AUPRC is computed using trapezoidal integration.

For the RAGTruth HOGwash/LUMINA comparison, we compute pointwise 95% confidence intervals using 2,000 paired bootstrap samples of responses. These intervals are not corrected for multiple comparisons. We do not report confidence intervals or significance tests for HDM-Bench.

C Implementation Details

C.1 Model Configurations

Source attribution uses Llama-2-7B-Chat, Llama-3.1-8B-Instruct, Qwen3-4B-Instruct-2507, and Mistral-7B-Instruct-v0.3. We use a maximum input length of 5,000 tokens for source attribution and 4,096 tokens for hallucination detection. Inputs exceeding these limits are excluded rather than truncated.

C.2 Input Construction

The source-attribution prompt uses the following plain-text structure rather than a native chat template:

```
Answer the question using only the
context below.  Answer concisely.

Context:
{ordered sources and titles}

Question:  {question}
Answer:
```

Method	RAGTruth		HDM-Bench	
	AUROC	AUPRC	AUROC	AUPRC
Negative mean log-probability	.517	.490	.631	.772
Token entropy	.694	.659	.734	<u>.855</u>
Context-removal gain	.590	.553	.688	.824
RAUQ	.656	.616	<u>.739</u>	.835
LLM-Check (attention)	.549	.552	.500	.708
LUMINA	<u>.744</u>	<u>.731</u>	.734	.847
HOGWASH	**.759**	**.740**	**.773**	**.885**

Table 5: **Contextual hallucination detection on RAGTruth and HDM-Bench.** Response-level AUROC and AUPRC using Llama-2-7B-Chat as the scoring model. RAGTruth includes 450 responses and HDM-Bench includes 190 responses. Bold and underlining mark the highest and second-highest values for each metric and dataset.

Retrieved sources appear before the question, allowing question positions to attend to source tokens. Source spans exclude instructions, question and response tokens, and special tokens. The complete reference response is evaluated under teacher forcing. Hallucination detection instead uses native chat formatting and complete generated responses.

C.3 Attention-Write Attribution

We compute signed source-specific Logit attributions using the attention weights, value vectors, and output projections of each attention head, as described in Section 3. Scores are summed across all heads, layers, and selected prediction positions.

C.4 GroundHOG Ranking

Signed token attributions are aggregated within each candidate source before normalization, as in Equation 6. Question and response source vectors are normalized separately by their L_1 norms and added. Summed Logit (question + response) instead uses the unnormalized sum, while question-only and response-only variants use their respective attribution vectors.

C.5 HOGwash Scoring

HOGwash measures the negative mean Logit attribution from the complete prompt to the response tokens. The prompt includes retrieved documents, instructions, and the question. Higher scores indicate greater hallucination risk. Unlike GroundHOG, HOGwash does not normalize separate source-attribution vectors.

D Baseline Implementations

D.1 Textual Retrieval and Entailment

We evaluate BM25, BGE (Xiao et al., 2024), and a DeBERTa-based NLI model (Sentence Transformers, n.d.). BM25 uses $k_1 = 1.5$ and $b = .75$. BGE uses BAAI/bge-base-en-v1.5 with cosine similarity, while NLI uses cross-encoder/nli-deberta-v3-xsmall with entailment scores. Retrieval variants use the question, response, or their combination to rank candidate passages.

D.2 Model-Internal Source Attribution

Attention baselines aggregate attention weights over the selected source and prediction positions. MIRAGE (Qi et al., 2024) identifies context-sensitive response tokens and attributes them to source tokens. AttnTrace (Wang et al., 2026) estimates source contributions from attention patterns. ALTI-Logit propagates attribution through earlier contextual mixing, while AttnLRP uses relevance propagation to input embeddings (LXT Contributors, n.d.). We evaluate response-only and combined question–response attribution where applicable.

D.3 Context-Perturbation Attribution

Leave-one-out attribution measures changes in response log-probability after removing individual sources. ARC-JSD (Li et al., 2026) measures changes in predictive distributions using Jensen–Shannon divergence. ContextCite estimates source contributions using randomized context ablations and a sparse linear model.

D.4 Hallucination Detection Baselines

We compare HOGwash against LUMINA (Yeh et al., 2026), negative mean log-probability, token entropy, context-removal gain, RAUQ, and LLM-Check (attention). LUMINA combines context-utilization and internal-knowledge signals; the other baselines measure uncertainty or context utilization.

E Additional Source Attribution Results

E.1 Answer-Localization and Retrieval Ablations

Table 6 reports question and response attribution ablations for MRQA and WikiQA. Table 7 compares the corresponding attribution targets using BGE retrieval scores.

E.2 Replacing the Question Signal with Retrieval Scores

We investigate whether question-side Logit attribution provides information beyond standard retrieval scores. We replace the question-side attribution in GroundHOG with BGE or BM25 scores while keeping the response-side Logit attribution unchanged. Both vectors are normalized by their source-level L_1 norms before addition. We also evaluate mean-centered BM25 scores to account for offsets introduced by normalization.

Table 8 shows that GroundHOG achieves higher AP than the retrieval-based combinations across all six dataset/model settings. These results suggest that question-side Logit attribution provides information not fully captured by lexical or embedding-based retrieval scores.

E.3 Question–Response Weight Sensitivity

We evaluate different weights for question and response attribution using $\lambda \in \{0, .25, .5, .75, 1\}$, where $\lambda = 0.5$ gives the same ranking as GroundHOG. Figure 3 shows that equal weighting achieves the highest model-averaged AP among the evaluated weights on both multi-hop benchmarks.

E.4 Attribution Propagation Comparisons

F Additional Qualitative Examples and Computational Requirements

F.1 Evidence Attribution Visualizations

Figure 4 compares Attention, Summed Logit, and GroundHOG on a multi-hop question. Underlining marks annotated supporting evidence, while colors indicate positive and negative attribution. The example illustrates how the methods distribute attribution across supporting passages.

F.2 Computational Requirements

Table 10 compares the number of model evaluations required by the attribution methods. GroundHOG uses a single forward evaluation without additional training or context perturbations. These counts do not account for differences in sequence length, memory usage, or computational overhead.

G Statistical Comparisons

We compare GroundHOG with response-only Summed Logit and the unnormalized combination of question and response attribution. Table 11 reports paired AP differences on HotpotQA and MuSiQue. Confidence intervals use 10,000 paired bootstrap samples. Significance is assessed using paired sign-randomization tests with Holm correction across the original 48-test ablation family.

GroundHOG improves AP over response-only attribution across all models and both datasets. Separate normalization provides smaller additional improvements, several of which remain significant after correction.

H AI Use Declaration

We used AI assistance to support manuscript drafting and editing, identify relevant literature, and assist with the execution of experiments. The research questions, methodological ideas, and experimental design were developed by the authors. All experimental results were independently verified by the authors, who remain responsible for the scientific claims, analyses, citations, and reported results.

Target / aggregation	Llama-2	Llama-3.1	Qwen3	Mistral
MRQA				
Question only	.519/.357	.542/.374	.530/.364	.527/.355
Response only	**.805/.710**	**.806/.713**	**.807/.717**	**.804/.711**
Question + response, unnormalized	.710/.582	.709/.579	.728/.611	.704/.573
GroundHOG	.776/.681	.781/.689	.785/.698	.782/.689
WikiQA				
Question only	.650/.471	.710/.548	.706/.557	.682/.507
Response only	**.969/.968**	**.969/.977**	**.970/.973**	**.974/.977**
Question + response, unnormalized	.945/.932	.958/.955	.964/.964	.957/.950
GroundHOG	.956/.959	.964/**.977**	.962/.964	.958/.964

Table 6: Answer-localization attribution ablations, AP/P@1. Each model–dataset column uses identical examples across the four variants. MRQA uses its 12-subset macro-average. Bold indicates the highest AP or P@1 within each model–dataset column, including ties.

BGE target / aggregation	HotpotQA	MuSiQue	MRQA	WikiQA
Question only	.646/.700	.620/.736	.513/.350	.753/.643
Response only	.577/.766	.571/.738	**.754/.642**	.945/.932
Question + response, unnormalized	.748/.830	.703/.812	.725/.599	.951/.937
Question + response, normalized	.745/**.836**	.701/**.814**	.735/.613	**.952/.941**
Concatenated question + response	**.753**/.796	**.713**/.794	.679/.548	.936/.914

Table 7: BGE target ablations, AP/P@1, across the four benchmarks. MRQA uses the subset macro-average.

Method	HotpotQA Llama-2	HotpotQA Llama-3.1	HotpotQA Qwen3	MuSiQue Llama-2	MuSiQue Llama-3.1	MuSiQue Qwen3
Summed Logit (question)	.634/.706	.692/.740	.679/.716	.570/.674	.626/.758	.641/.744
Summed Logit (response)	.619/.800	.661/.852	.621/.830	.635/**.910**	.671/.954	.655/.944
BGE question	.645/.701	.645/.701	.645/.701	.624/.744	.624/.744	.624/.744
BM25 question	.553/.639	.553/.639	.553/.639	.473/.518	.473/.518	.473/.518
BGE q + Logit r	.683/.809	.727/.860	.696/.829	.715/.908	.763/.953	.758/.945
BM25 q + Logit r	.706/.807	.739/.866	.719/.837	.711/.904	.740/.949	.733/.945
Centered BM25 q + Logit r	.721/.819	.748/.864	.729/.848	.717/.898	.743/.949	.735/.945
GroundHOG	**.779/.842**	**.825/.884**	**.808/.874**	**.772**/.908	**.821/.962**	**.824/.950**

Table 8: Query-relevance replacement controls, reporting AP/P@1. Bold and underlining mark the highest and second-highest values, including ties. Each combination retains the same response Logit attribution and replaces the question-side scores.

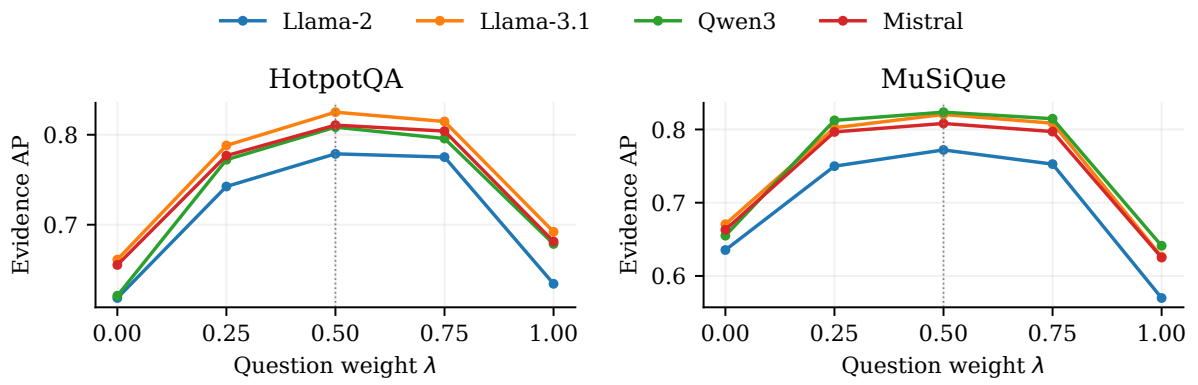


Figure 3: Evidence AP as the question weight varies. Zero uses response attribution alone, one uses question attribution alone, and the dotted line marks GroundHOG.

Method	Llama-2	Llama-3.1	Qwen3	Mistral
HotpotQA				
ALTI-Logit, response	.604/.712	.593/.678	.575/.672	.607/.680
ALTI-Logit, normalized q+r	.618/.626	.585/.546	.563/.554	.603/.584
AttnLRP, response	.655/.824	.699/.868	.679/.866	.710/.866
AttnLRP, unnormalized q+r	.790/.840	.832/.880	.806/.834	.813/.850
AttnLRP, normalized q+r	**.793/.858**	**.839/.900**	**.820/.896**	**.827/.890**
MuSiQue				
ALTI-Logit, response	.595/.822	.621/.862	.604/.856	.625/.880
ALTI-Logit, normalized q+r	.606/.742	.616/.752	.600/.740	.616/.744
AttnLRP, response	.644/**.932**	.656/.958	.627/**.958**	.668/**.962**
AttnLRP, unnormalized q+r	.737/.871	.798/.893	.769/.875	.799/.886
AttnLRP, normalized q+r	**.736**/.918	**.818/.960**	**.779**/.956	**.808**/.954

Table 9: Attribution-target comparisons on HotpotQA and MuSiQue, AP/P@1. ALTI-Logit and AttnLRP propagate attribution to input tokens. Normalized q+r separately normalizes question and response source scores before addition. Bold indicates the highest values within each model and dataset among the displayed methods.

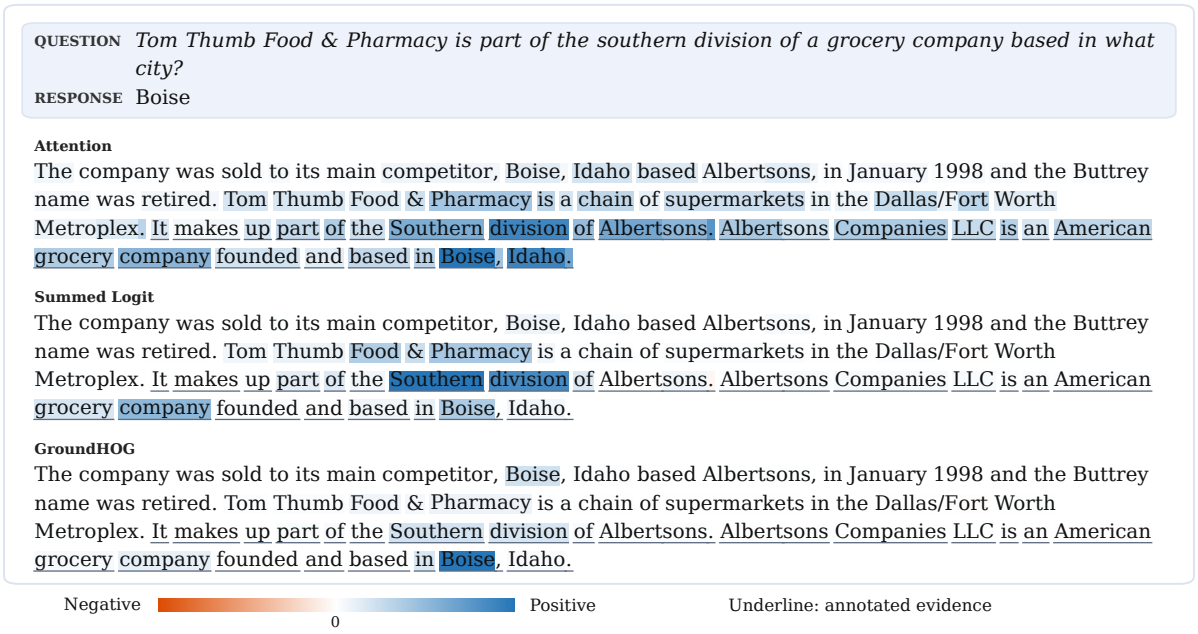


Figure 4: Complementary evidence linking Tom Thumb to its parent company and headquarters. Attention, Summed Logit, and GroundHOG are compared on the same source text and response. Underlining marks annotated supporting evidence. Colors indicate signed attribution, with independent scaling for each method.

Method	Forward passes	Backward passes
GroundHOG / Summed Logit / Attention	1	0
ALTI-Logit	1	0
AttnLRP (question + response)	1	2
Gradient saliency	1	1
MIRAGE	2	Multiple
AttnTrace-1	1	0
AttnTrace	30	0
Leave-one-out / ARC-JSD	$1+K$	0
ContextCite	32	0

Table 10: Computational requirements of source-attribution methods. K denotes the number of candidate sources. Pass counts exclude additional processing and do not directly reflect runtime or memory usage.

Dataset	Model	vs. Response only ΔAP [95% CI]	vs. Unnormalized q+r ΔAP [95% CI]
HotpotQA	Llama-2	+.160 [.143, .178]*	+.012 [.001, .023]
HotpotQA	Llama-3.1	+.164 [.146, .182]*	+.025 [.015, .036]*
HotpotQA	Qwen3	+.187 [.169, .206]*	+.010 [-.001, .021]
HotpotQA	Mistral	+.155 [.138, .173]*	+.021 [.008, .034]*
MuSiQue	Llama-2	+.137 [.121, .153]*	+.021 [.014, .029]*
MuSiQue	Llama-3.1	+.150 [.133, .166]*	+.021 [.014, .029]*
MuSiQue	Qwen3	+.169 [.152, .186]*	+.007 [.002, .013]
MuSiQue	Mistral	+.145 [.129, .161]*	+.016 [.008, .024]*

Table 11: GroundHOG AP improvements over response-only attribution and unnormalized question–response attribution. Positive values favor GroundHOG. Stars indicate Holm-adjusted $p < .05$ within the original 48-test family. Confidence intervals are pointwise.